FUEL MOBILITY SERVICE

Product Planning Document · v1.0

EMERGENCY FUEL & ROADSIDE MOBILITY PLATFORM

Prepared for: Stakeholders, Investors & Engineering Teams

Classification: Confidential · 2025

<45 min
Dispatch-to-Arrival (Urban Pilot)

99%
Live Location Update Reliability

100%
SOS-Enabled Safety Mode Trips

10k
Concurrent Tracking Sessions / Region

1 PRODUCT VISION & STRATEGY

Mission Statement

Deliver safe, fast, and traceable emergency fuel and roadside mobility assistance — especially for stranded drivers and women travelers — through a technology-first, operations-backed platform that prioritises trust, speed, and compliance.

Value Proposition

Stakeholder	Core Value
Customers	One-tap emergency fuel + live ETA + trusted agents + safety features
Operators	Dispatch, fleet, compliance, and fraud controls in one admin stack
Partners (B2B)	API-ready fleet and payment integrations for hotels, fleets, and insurers

Strategic Goals — 12-Month Targets

Goal	Success Metric
Emergency Fuel Delivery	Median dispatch-to-arrival < 45 min (urban pilot)
Real-Time Tracking	99% location updates every 10–15s during active trip
Women Safety	100% optional safety mode trips with live share + SOS
Fast Delivery SLA	90% on-time SLA for standard orders
Secure Operations	Zero payment disputes unresolved >48h; OTP on 100% deliveries
Scalable Architecture	Support 10k concurrent tracking sessions per region

Target User Segments

Segment	Persona	Primary Need
B2C Drivers	Stranded commuter, highway traveler	Emergency fuel, SOS, tracking
Women Drivers	Safety-conscious urban/suburban user	Verified agents, trip share, SOS
Fleet / Logistics	Fleet manager, SMB operator	Bulk orders, agent assignment, reports
Operations	Admin, support, compliance	Dispatch, disputes, audit trails
Field Agents	Fuel delivery agent	Job queue, navigation, proof-of-delivery

2 MVP PLANNING & ROADMAP

MUST-HAVE (MVP — Launch-Ready)

- Phone/email OTP auth + role-based login (customer, agent, admin)
- Emergency + standard fuel request (location, fuel type, quantity estimate)
- Agent assignment — manual + nearest-available rule
- Live GPS tracking (customer + admin map)
- Live ETA (routing API + status machine)
- Razorpay/Stripe-style payment + cash-on-delivery flag
- Delivery OTP verification + photo proof
- Push/SMS notifications across request lifecycle
- Basic admin dashboard (orders, agents, status override)
- Women safety mode v1: trusted contacts share link + SOS button
- Audit logs for safety and payment events

MVP+ (Optional)

- Scheduled fuel delivery (non-emergency)
- In-app chat (customer ↔ support)
- Agent ratings and tipping
- Promo codes / wallet credits
- Multi-language (EN + one regional language)

POST-MVP / v2+

- AI demand forecasting and dynamic pricing
- Predictive agent pre-positioning
- B2B API for insurers and roadside assistance partners
- EV charge-assist / battery boost partnerships
- Subscription plans (fleet, premium safety)
- Computer-vision fuel gauge estimation from photo

Phase Roadmap (~26 Weeks)

Phase	Duration	Focus
0 — Discovery	2 weeks	Compliance review, wireframes, pilot city selection
1 — Foundation	4 weeks	Users, auth model, admin concepts
2 — Core Orders	4 weeks	Orders, pricing, payments, lifecycle
3 — Field Ops	4 weeks	Agent workflows, tracking, ETA
4 — Safety & Trust	3 weeks	Women safety, SOS, OTP proof
5 — Ops & Polish	3 weeks	Admin, support, fraud rules

Phase	Duration	Focus
6 — Pilot Launch	2 weeks	Beta, runbooks, App Store / Play Store readiness
7 — Post-Launch	Ongoing	AI features, B2B API, multi-city expansion

3 USER ROLES & MODULES

The platform is structured around five distinct roles, each with dedicated surfaces and a scoped RBAC permission model supporting multi-city tenancy.

Role	Primary Surfaces	Core Modules
Customer	iOS / Android / PWA	Auth, fuel request, emergency, track, pay, safety mode, SOS, history
Fuel Delivery Agent	Mobile App	Auth, job inbox, navigate, status updates, OTP scan, incident report
Fleet Manager	Web Dashboard	Agent roster, vehicle/fuel inventory, zones, performance KPIs
Admin	Web Console	Users, orders, dispatch override, pricing, compliance, fraud flags
Support Team	Web + Ticketing	Tickets, refunds, SOS escalation, chat/call logs

Permission model: RBAC with scopes (customer, agent, fleet_manager, admin, support) + optional org/tenant for multi-city operations.

4 FEATURE BREAKDOWN

Feature	Description
4.1 Authentication	Phone OTP (primary) + email optional. JWT access tokens (15 min) with 7-day refresh and device binding. KYC flags for agents covering ID, license, and background check status. Session revocation on password/device change.
4.2 GPS Tracking	Agent foreground location stream at 10–15s intervals during active jobs. Customer map polyline + agent marker via WebSocket or SSE. Geofence arrival radius (~100m) triggers 'agent nearby' prompt. Raw points stored in time-series; aggregated for billing and dispute resolution.
4.3 Emergency Request	Priority queue: EMERGENCY > STANDARD. Auto-collects GPS, vehicle type, fuel type, phone, and optional photo. SLA timer with auto-escalation to support if unassigned within N minutes. Surge pricing rules (transparent, capped).
4.4 Fuel Ordering	Catalogue: petrol/diesel/CNG (region-dependent), min/max liters. Full price breakdown: fuel + delivery fee + tax + surge. Order states: draft → placed → assigned → en_route → arrived → delivering → completed/cancelled.
4.5 Live ETA	Google Maps / Mapbox Directions API integration. Recalculated on agent movement and traffic updates. Displayed as range (e.g. 12–18 min) to reduce false precision.
4.6 Payment Integration	Gateway: Razorpay (IN) or Stripe (global) via Payment Intents. Methods: UPI, card, wallet, COD with agent confirmation. Webhooks for payment.captured, failed, and refunded events. Idempotency keys on all payment creation.
4.7 Delivery Management	Dispatch rules: nearest online agent → load balance → emergency-certified skill. Agent accept/decline with timeout. Proof: OTP + optional photo of dispensing. Partial delivery / quantity adjustment with customer approval.
4.8 Notifications	Channels: FCM/APNs push, SMS, email receipts. Event triggers: assigned, en_route, arrived, OTP, payment, SOS. User preferences + quiet hours (except emergency/SOS overrides).
4.9 Women Safety Mode	Opt-in profile flag + per-trip toggle. Trusted contacts receive live trip link showing map, agent profile, and vehicle. Periodic 'I'm OK' check-ins; missed check-in triggers nudge then escalation. Night mode: enhanced logging + mandatory agent photo ID visible in app.

Feature	Description
4.10 SOS System	Long-press or volume key combo trigger. Payload includes location, order ID, agent ID, and recording consent flag. Notifies trusted contacts + support + optional local emergency API (where legal). Immutable sos_events log with post-incident review queue for admins.

5 USER FLOWS

Customer Journey

1. Onboard → verify phone → add vehicle(s) → optional trusted contacts

2. Home → 'Emergency Fuel' or 'Order Fuel' → confirm location & fuel type

3. Pay (gateway / COD) → track live on map → receive OTP from agent

4. Delivery confirmed → receipt generated → rate agent

Delivery Workflow (Agent)

1. Go online → receive job notification → accept (or auto-assigned)

2. Navigate to customer → update status to en_route → geofence arrival

3. Verify customer identity → dispense fuel

4. Enter liters dispensed → customer confirms OTP → optional photo proof → mark completed

5. Offline sync if network loss during delivery

Emergency Workflow

1. Customer taps Emergency → priority=EMERGENCY flag set

2. Shorten assignment timeout; notify on-call support immediately

3. If no agent assigned in 5 min → expand radius + notify fleet manager

4. SOS during trip → freeze reassignment; open war room ticket

5. Post-event: mandatory admin review + customer follow-up required

Admin Workflow

1. Dashboard: live map of active orders + SOS alerts

2. Manual assign/reassign agent; cancel orders with applicable refund rules

3. Review KYC, agent documents, and safety incidents

4. Configure pricing zones, surge caps, and service areas

5. Export reports: orders, revenue, SLA performance

6 DATABASE & API PLANNING

Database Architecture

Recommended stack: PostgreSQL (OLTP) + PostGIS (geospatial) + Redis (cache/sessions) + optional time-series store for location_points.

Table	Key Fields
users	id, phone, email, name, safety_mode_enabled, status
agent_profiles	user_id, license_no, kyc_status, safety_verified, lat/lng, is_online
orders	customer_id, agent_id, type, priority, status, pickup_lat/lng, fuel_type, liters, price_total, safety_mode
payments	order_id, gateway_id, amount, status, method, idempotency_key
delivery_proofs	order_id, otp_hash, verified_at, photo_url
location_points	order_id, agent_id, lat, lng, recorded_at
sos_events	order_id, user_id, lat, lng, triggered_at, resolved_at
pricing_zones	geo_polygon, base_fee, per_km, surge_cap
audit_logs	entity_type, entity_id, action, actor_id, payload

API Design

Style: REST v1 + WebSocket for live tracking. Base URL: https://api.fuelmobility.com/v1 Response envelope: { data, meta, errors[] }

Domain	Example Endpoints	Access
Auth	/auth/otp/send, /auth/otp/verify, /auth/refresh	Public
Profile	/me, /me/vehicles, /me/trusted-contacts	Customer
Orders	POST/GET /orders, /orders/{id}/cancel	Customer
Emergency	POST /orders/emergency	Customer
Tracking	GET /orders/{id}/tracking, WS	Customer, Agent, Admin
Agent Jobs	/agent/jobs, accept, status updates	Agent
Delivery	verify-otp, proof upload	Agent + Customer
Safety	/orders/{id}/sos, /trips/{share_token}	Customer / Public share
Admin	/admin/orders, agents, dispatch	Admin, Support
Fleet	/fleet/agents, reports	Fleet Manager

7 SECURITY & SCALABILITY

Security Controls

Area	Approach
JWT Auth	RS256; short-lived access tokens; refresh rotation; blacklist on logout
Encryption	TLS 1.3 in transit; AES-256 at rest for PII; OTPs hashed with Argon2
Payments	PCI compliance via gateway tokenisation; webhook HMAC signature verification
Fraud Prevention	Device fingerprinting, velocity limits, GPS sanity checks on orders
Delivery OTP	6-digit, 10-min TTL, max 3 attempts, single-use enforcement
API Security	Rate limiting, WAF, OWASP ASVS L2 target compliance
Safety Logs	SOS events append-only; admin overrides fully audited
Compliance	Fuel transport licences per state; GDPR-style data export/delete

Scalability Phases

Phase	Architecture	Key Decisions
Phase 1 — MVP	Modular monolith	Single API + worker; PostgreSQL + Redis + object storage for delivery proofs
Phase 2 — Growth	Extract hot paths	ALB/nginx → API gateway → core + dedicated tracking service; Redis queue for notifications and SOS
Phase 3+ — Scale	Microservices	Split auth, orders, payments, tracking, notifications, safety when traffic and team justify

Technology Reference Options

Layer	Options
Mobile	React Native · Flutter
Web Admin	Next.js
API / Backend	NestJS · FastAPI
Database	PostgreSQL + PostGIS
Cache / Queue	Redis + BullMQ / SQS
Maps	Google Maps · Mapbox
Payments	Razorpay (IN) · Stripe (Global)
Cloud	AWS · GCP | Docker + Terraform + OpenTelemetry

8 RISKS & OPEN DECISIONS

Risk Register

Risk	Mitigation
Fuel transport regulations vary by state	Legal review per state prior to launch; licensed and compliant agents only
SOS liability exposure	Clear T&C; partner with official emergency services where possible
Agent supply shortage in pilot city	Fleet onboarding incentives and referral programme
GPS spoofing by agents	Cross-check speed, network cell-tower data, and geofence consistency
Payment fraud	3D Secure; hold payout to agent until delivery confirmed via OTP

Planning Decisions to Finalise

#	Decision	Context
1	**Pilot Geography & Fuel Types**	Confirm target city, and which fuel types to launch — petrol / diesel / CNG.
2	**Payment Provider**	Confirm primary market to select between Razorpay (India) and Stripe (global rollout).
3	**SOS & Safety Legal Model**	Define liability framework and emergency service partnership agreements.
4	**Go-to-Market Sequencing**	Decide whether to launch B2C first and add fleet B2B, or pursue fleet B2B from day one.
